Ankush Sonawane

Project Engineer, AI/ML & HPC · LLM Infrastructure · MLOps

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SUMMARY

AI/ML engineer with 2.5 years at C-DAC building AI infrastructure on India's national HPC systems (PARAM Shavak, PARAM Rudra). Shipped a self-service model serving platform, the 14+ GPU container stack for the PARAM Shavak 2.0 product, and fully local LLM and RAG systems. Delivered 10+ workshops and faculty programmes on LLMs, distributed deep learning and HPC across India.

TECHNICAL SKILLS

LLMs and GenAI: vLLM, SGLang, llama.cpp, LangChain, LangGraph, LiteLLM, Hugging Face, RAG, LoRA / QLoRA
ML and deep learning: PyTorch, TensorFlow, scikit-learn, RAPIDS, YOLO, Transformers, CNNs, GANs, DDP / FSDP
Containers and MLOps: Apptainer, Podman, Docker, Kubernetes (fundamentals), OCI registries, MLflow, DVC
HPC: Slurm, Spack, CUDA, NCCL, OpenMP offloading, Lustre, multi-node GPU benchmarking
Services and access: FastAPI, Gradio, Keycloak, LDAP, Open OnDemand
Languages and OS: Python, C++, Bash, SQL (MySQL, PostgreSQL), Linux

EXPERIENCE

Project Engineer · Centre for Development of Advanced Computing (C-DAC), Pune — Apr 2024 – Present

AI-Factory Model Hub: self-service model serving platform — 2026 – Present

- Architected Model Hub, which lets cluster users train, register and deploy models as GPU endpoints in one click; shipped v1 and v2 and now building v3 for the Rudra AI rack (~100 users, fully offline).
- Linked MLflow model registration to one-click deployment of ML, DL and LLM models as Slurm-scheduled GPU endpoints, writing their Kubernetes deployment manifests; all three endpoint types passed on A30 GPUs.
- Secured access with Keycloak SSO over LDAP and an HTTPS gateway using hashed per-user API keys and sharing controls; found and closed 2 gateway bypasses and added 15 unit tests.
- Automated installation as 9 idempotent modules driven by one config file (100+ commits) and wrote 10 user and admin guides covering login, containers, MLflow, Model Hub and API keys.

AI container stack — 2025 – Present

- Built and maintained 14+ Apptainer GPU containers (PyTorch, TensorFlow, RAPIDS, vLLM, SGLang, llama.cpp, MLflow) on CUDA 12.9 with pinned dependencies, shipped standalone with the PARAM Shavak 2.0 product.
- Reused the same containers in AI-Factory on Rudra as training environments and as the base for deployment OCI images, so training and serving run in identical environments.
- Built the containers from scratch on AlmaLinux in place of NGC base images, and sped up builds with uv, host-side caching and nearest-mirror installs.
- Wrote a Bash test-automation script with a standard test suite per container, covering GPU visibility, package checks, and multi-node training and serving runs.
- Exposed the same containers as kernels in AI-Factory's Jupyter tool, giving users seamless access to the exact training environments.
- Set up a rootless Podman build and local OCI registry pipeline for distributing images across the AI-Factory cluster.

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EXPERIENCE, CONTINUED · C-DAC

LLM systems and benchmarking

2025 – Present

- Engineered C-Nidan, an LLM-based HPC log-analysis system: built the log ingestion and inference pipeline and a one-click installer (Conda, Apptainer + vLLM via Spack, health checks) validated on 3 national clusters.
- Designed a fully local, vectorless RAG system on PageIndex with vLLM, LiteLLM and FastAPI, including vision-based indexing of scanned PDFs; stabilised throughput by serialising requests to end KV-cache contention.
- Benchmarked LLM inference (Llama 3 8B; Qwen 3.5 0.8B to 35B-A3B; Q4 to F16) with llama-bench, LLMPerf and vLLM bench, and developed a GUI to run and compare benchmarks.

Deep learning and HPC research

2024 – 2025

- Trained YOLOv8/YOLOv11 lung-tumour detectors on multi-GPU nodes after building DICOM-to-PNG and VOC-to-YOLO data pipelines (Lung-PET-CT-Dx); deployed retinal AV-segmentation and YOLO apps on PARAM Shavak with Gradio UIs.
- Scaled PyTorch training across multi-node GPU clusters with DDP and Slurm, and ran CycleGAN multi-node for a physics-informed GAN (PI-GAN) research project.
- Benchmarked NAMD, GROMACS, LAMMPS, WRF, OpenFOAM and HPL on PARAM Shivay and Shavak; compiled OpenFOAM and ParaView from source on PARAM Utkarsh and IIT Goa's ParamVidya.
- Delivered an LLM resume-screening system (LLaMA 3.1 8B, Gradio) for PDF, DOCX and TXT files, and LLM endpoints that predict job domain and explain Slurm errors for the HPC monitoring portal.

Teaching, mentoring and knowledge transfer

2024 – Present

- Authored 100+ hours of Data Science, ML and DL course material (3-day to 40-hour tracks) and LLM content on Transformers, attention and LoRA / QLoRA; prepared and reviewed NPTEL assessment questions.
- Mentored multiple UG interns end to end, from selection interviews through project scoping and weekly reviews to certification.
- Ran knowledge-transfer sessions on distributed deep learning and AI containers, and onboarded new joinees to Slurm and Spack.

WORKSHOPS, TOURS AND OUTREACH

- **KL University, Hyderabad:** one-week faculty development programme (2026)
- **NIT Arunachal Pradesh:** one-week HPC workshop with hands-on sessions (2026)
- **NESAC, Shillong:** one-week workshop on distributed deep learning (2026)
- **CHARUSAT University:** workshop on large language models (2026)
- **Pimpri Chinchwad University:** two-day AI/HPC training with hands-on labs (2026)
- **Impact AI Summit and C-DAC Conclave:** represented PARAM Shavak and BOSS OS with live demos (2026)
- **SCI event:** technical session on deep learning Apptainer containers (2026)
- **IIT (BHU) Varanasi:** session on distributed deep learning training
- **Tanzania cohort and Govt. Polytechnic College:** hands-on ML and AI/ML sessions (2025)
- **AICTE IIT Bombay HPC Workshop:** lab and exam support, Pune Darshan tour logistics (2024)
- **Demos and awareness:** presented PARAM Shavak to the Sarvam AI team and PCMC university officials; led C-DAC's HPC awareness programme for schools.

EDUCATION

PG Diploma in Artificial Intelligence · C-DAC, Innovation Park, Pune

B.E. in Computer Engineering · Savitribai Phule Pune University, Pune